BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Mulitple Choice Questions (MCQ's)

> Note: Q.1: Choose the correct answer for each form th

Q1. A point where a function 'f' changes form an increasing function to decreasing function is called:

(a) A point of maximum value
(b) A point of minimum value
(c) point of zero value
(d) None of these

Q2. ∫x cos x dx

(a) x sin x + cosx + c
(b) x sin x + cos x + c
(c) x sion x − cos x + c
(d) x cos x − sin x + c

Q3. If (x1, y1) and (x2 − y2) are end point of the diameter then centre of circle is:

(a) (x1−x2)/2, (y1−y2)/2
(b) (x1+x2)/2, (y1+y2)/2
(c) (x1−x2)/2, (y1−y2)/2
(d) (x1 + x2, y1 + y2)

Q4. If P(x1, y1) and Q(x2, y2) are any two points on parabola $y^2 = 4ax$ then eq: of the chord PQ will be:

(a) (y + y1)(y1 − y2) = 4a(x − x1)
(b) (y − y1)(y1 + y2) = 4a(x − x1)
(c) (y + y1)(y1 + y2) = 4a(x − x1)
(d) (y + y1)(y1 + y2) = 4a(x + x1)

Q5. The vector product of a vector by itself is always:

(a) 1
(b) -1
(c) greater than 1
(d) 0

Q6. lim (x→0) (1 + 3x)^{2/3}/x

(a) e^3
(b) e^{-6}
(c) e^6
(d) e

Q7. The point (-3,0) (0,0) and (5,0) are:

(a) vertices of triangle
(b) collinear point
(c) Non-collinear
(d) None

Q8. The x-intercept of the line $3x + 4y - 7 = 0$

(a) -7/3
(b) 7/3
(c) 7/4
(d) -7/4

Q9. The slope of the curve $y = 3e^{2x}$ at $x = 0$ is equal to:

(a) 3
(b) 6
(c) 5
(d) 2

Q10. Maximum values of f(x) are minimum value of:

(a) f (x)
(b) -f
(c) f'(x)
(d) -f'(x)

Q11. $\int(ax + b)^n dx = ?$

(a) $(ax+b)^{n+1}/(an+1)$
(b) $(ax+b)^{n-1}/(an+1)$
(c) $(ax+b)^{n+1}/(an+1)$
(d) $(ax+b)^{n+1}/(a(n+1))$

Q12. Two circles are said to be concentric if they have:

(a) same radii
(b) same centre
(c) same diameter
(d) different centre

Q13. The line $y = mx + c$ is tangent to ellipse $x^2/a^2 + y^2/b^2 = 1$ if

(a) $c = a/m$
(b) $c^2 = a^2m^2 + b^2$
(c) $c^2 = a^2m^2 - b^2$
(d) $c^2 = a^2(1+m^2)$

Q14. What does scalar triple product gives:

(a) Area of square
(b) Volume of cylinder
(c) Volume of square
(d) Volume of parallelepiped

Q15. If $\lim (a^x - 1)/x = ?$

(a) lna
(b) ln x
(c) aln x
(d) 1/a ln a

Q16. If two non vertical lines are parallel their inclinations and their slopes are:

(a) Perpendicular
(b) Zero
(c) Not equal
(d) Equal

Q17. Which one is two intercept form:

(a) x/a + y/b
(b) x/a − y/b
(c) x/a + y/b = 0
(d) x/a + y/b = -1

Q18. (-2, 5), (4, 1) and (5, 4) are the vertices f triangle its centroid is:

(a) (7/3, 8/3)
(b) (11/3, 8/3)
(c) (7/3, 10/3)
(d) (0,1)

Q19. If ∫πdθ = ?

(a) π + c
(b) πθ + c
(c) θ + c
(d) π/3 + c

Q20. ∫[f(x)]^n f'(x) dx = ?

(a) [f(x)]^{n+1}/(n+1)
(b) ln [f(x)]^{n+1}
(c) ln [f(x)]^n
(d) All of these

Q21. The general equation of he circle whose centre lies on x-axis:

(a) $x^2 + y^2 + 2fy + c = 0$
(b) $x^2 + y^2 - 2gx + c = 0$
(c) $x^2 + y^2 + 2gx + c = 0$
(d) $x^2 + y^2 + 2gx - c = 0$

Q22. The point P(x1, y1) lies outsides the circle x2 + y2 + 2g x + fy + c = 0 if

(a) $x1^2 + y1^2 + 2gx1 + 2fy1 + c > 0$
(b) $x1^2 + y1^2 + 2gx1 + 2fy1 + c < 0$
(c) $x1^2 + y1^2 + 2gx1 + 2fy1 + c = 0$
(d) $x1^2 + y1^2 + 2gx1 + 2fy1 + c < 0$

Q23. If a = (2, -1, 2) B = (-4, 1, -1) then AB = ?

(a) -6i − 2j + 3k
(b) -6i + 2j + 3k
(c) 6i + 2j − 3k
(d) 6i − 2j − 3k

Q24. lim (x → ∞) [illegible in source scan] e^x = ?

(a) 1
(b) 0
(c) ln x
(d) ∞

Q25. Two lines 5x − 3y + 8 = 0 and 10x − 6y − 7 = 0 are

(a) Parallel
(b) Perpendicular
(c) Coincident
(d) None of these

Q26. Lim (x→0) $(x^2 - 1)/(x^2 - x)$

(a) 1
(b) 2
(c) x
(d) ∞

Q27. The equation of the x −axis itself as

(a) x = 0
(b) y = 0
(c) y = x
(d) None of these

Q28. Three lines are concurrent of |a1 b1 c1; a2 b2 c2; a3 b3 c3| = ?

(a) 1
(b) -1
(c) ±
(d) 0

Q29. If the point on the graph y = f(x) where f(x) is neither increasing nor decreasing and f(x) = 0 then that point is called;

(a) Maximum point
(b) Minimum point
(c) Stationary point
(d) Turning point

Q30. I d/dx [φ(x)] = f(x) then function Θ(x) is called:

(a) Derivative of f(x)
(b) Antiderivative of f(x)
(c) inverse of f(x)
(d) Definite integral

Q31. The centre of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$,

(a) (g, -f)
(b) g, f)
(c) (g, f)
(d) (g, -f)

Q32. ∫f(x) dx ∀ a,b ∈ R, a ≤ b is called:

(a) Definite integral
(b) Indefinite integral
(c) Simple integral
(d) None

Q33. If a = [P, 1, 0], b = [1, 1, 3] and c = [2, 1, -2] are coplanar then P = ?

(a) 3/4
(b) 4/5
(c) 6/5
(d) 8/5

Q34. If the point p (x, y), x is called.

(a) ordinate
(b) abscissa
(c) pole
(d) None of these

Q35. If a ≠ 0, c ≠ 0 and b = 0 then ax + by + c = 0 represents a line:

(a) Parallel to x −axis
(b) Parallel to y −axis
(c) Passing through the origin
(d) bisecting 1st and 3rd quad

Q36. $f(x) = x^3 + 4x^2 + 3x + 9$ then f''(x) = ?

(a) 6x + 8
(b) $6x^2 + 8x$
(c) 6x − 4
(d) 3x + 9

Q37. Two non-zero vectors U and V are parallel if:

(a) u : v = 0
(b) u x v = 0
(c) both (a) and (b)
(d) None of these

Q38. If we put x = a + h in function f(x) then the limit of f(x) as x → a is:

(a) the limit of f(x) as h → 0
(b) limit of f(a + h) as x → a is
(c) the limit of f(a + h) as h → a
(d) limit of f(a + h) as h → 0

Q39. y − co-ordinate of any point on the x-axis

(a) zero
(b) one
(c) y
(d) x

Q40. The point of concurrency of there lines 2x − 5y = 0, 3x − 17y = 0 and x + 6x = 0 is

(a) (-1, -1)
(b) (0, 0)
(c) (1, 1)
(d) 1/2, 1/2

Q41. The lines lying on the same plane are called:

(a) Collinear lines
(b) coplanar lines
(c) Concurrent lines
(d) Coincident lines

Q42. If y = e^{2x} -1 then dy/dx is euqla to:

(a) 2e^{2x} − 1
(b) 2e^{2x}
(c) 2ex
(d) None of these

Q43. The function f(x) = sin x as maximum value:

(a) 0
(b) 1
(c) -1
(d) -∞

Q44. The inverse operator of d/dx is

(a) ∫ dx
(b) dx
(c) Δ^2
(d) none of these

Q45. If the centre of circle $x^2 + y^2 + ax + by + 2 = 0$ is at (4, -8) the a + b = ?

(a) -4
(b) 4
(c) 8
(d) -8

Q46. The centre of circle at origin and radius r, h = 0, k = 0 then

(a) $x^2 - y^2 - r^2 = 0$
(b) $x^2 + y^2 = r^2$
(c) $y^2 - x^2 = r^2$
(d) $x^2 - y^2 = -r^2$

Q47. The unit vector I is:

(a) [0, 0, 1]
(b) 0, 1, 0]
(c) 1, 0, 0]
(d) None of these

Q48. If $p(\theta) = (x, y)$ when $\theta = 180°$ then $(x, y) =$

(a) (0, 1)
(b) (0, 1)
(c) (-1, 0)
(d) (1, 0)

Q49. Find θ, if s = 20 cm and r = 2m

(a) 0.1 radian
(b) 10 radian
(c) 0.1 degree
(d) 10 degree

Q50. $\cos 2\theta =$ _____

(a) $2\cos^2\theta$-1
(b) $\cos^2\theta - \sin^2\theta$
(c) $1 - 2\sin^2\theta$
(d) All of these

Section-B

> Note: Solve any SIX of the following questions. Each question carries 05 marks:

Q2. Find the limit of $\lim_{x\to 0} (1 - \cos 2x)/(1 - \cos 3x)$

Q3. If $f(x) = (\tan^{-1}x, \cos x + \sin x)$ find f(x).

Q4. Find the equation of a line through the intersection of the lines $5x - 6y = 1$, $3x + 2y + 5 = 0$ and perpendicular to $5y - 3x - 11 = 0$

Q5. Fine the extreme value of the function f given by $f(x) = x (x -1) (x -2)$ for all $x \in R$.

Q6. Evaluate $\int (3x + 2)/(x - 1)\, dx$

Q7. Find the equation of the circle having (-5, 6) and (3, -4) the end points of a diameter.

Q8. Find the equation of the tangent and normal to the hyperbold $2x^2 - 3y^2 = 1$ and at the point (2, 1)

Q9. The x − intercept of a line is the reciprocal of its y − intercept, The line passes through (2, -1). Find its equation.

Q10. If a function, given by $y = f(x)$, $\sqrt{(x^2 + y^2)} = \ln (x^2 - y^2)$ find dy/dx, where if exist.

Section-C

> Note: Solve any TWO of the following question.

Q11. (a) If y − tan2x, find dy/dx from definition. (b) Evaluate [2i + 3j − 1, 3k, j]

Q12. (a) The measure of the angle form a line l1 with slope -1/2 to a line l2 is 3π/4. Find the slope of l2. (b) Find ∫x² e^x dx,

Q13. (a) If y = f(x) = ae^x − be^{2x} + ce^{3x} for all x ∈ R wher a, b, c ∈ R, show that d³y/dx³ − 6d²y/dx² + 11dy/dx − 6y = 0 (b) Find lim (x→∞) √(x² − 4)/(x² + 1)^{1/2}